The Base-Derivative of the Kneser Family: Selection Principle, Uniqueness, and Existence via a Riemann–Hilbert Problem

Paper V of the mixed-base tetration series

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Abstract

Let $A_b = \mathrm{slog}_b$ be the regular (Kneser/Paulsen–Cowgill) superlogarithm family, $\beta = \ln\ln b$, and let $V = \partial_\beta A_b|_{b=d}$ be its base derivative (the response kernel of the mixed-base phase law). We record the selection principle that characterizes V: the cohomological equation $V(d^y) - V(y) = -A_d'(d^y)\, y\, (\ln d)\, d^y$, reality on $\mathbb{R}$, the prescribed fixed-point singularity with pole coefficient $dL/d\beta = \ln d\, L^2/(1-\lambda)$ and logarithmic coefficient $d\lambda/d\beta$, and the normalization $V(1) = 0$. We prove *uniqueness* under these conditions, reduce *existence* to a scalar Riemann–Hilbert (conjugate-function) problem for the β-derivative of the Kneser theta map, and then *solve* that boundary problem. The solution rests on three elementary structural facts established here: the geometric field $G := \partial_\beta\chi/\chi'$ built from the Koenigs data satisfies the *exact cocycle* $G(d^y) = (\ln d)\, d^y\, (G(y) - y)$, so that $A_d' G + h \circ A_d$ solves the cohomological equation for *every* 1-periodic h; the boundary data $u = \mathrm{Im}\big[(A_d' G) \circ T_d\big]$ is 1-periodic, extends *continuously* to the closed circle with $u(\mathbb{Z}) = 0$ and Dini modulus $O(\delta \ln^2(1/\delta))$ — the $\ln\ln$ cascade of the Koenigs coordinate at the punctures is annihilated by the collapse $|1/\chi'| = O(\delta\ln(1/\delta))$; and the one-sided limits of the geometric field at the punctures are the exact orbit values $V(T_d(m))$, e.g. $-e/T_e'(1)$ at $m = 1$ for $d = e$ (verified to ten digits). Consequently the conjugate-function step is classical (Privalov–Dini; no weighted class is needed), and interior recovery follows by Schwarz reflection, Riemann removability at the punctures, and the $\mathcal{S}_\beta/\mathcal{S}$ pole expansion at the fixed point. The outcome is an existence-and-uniqueness theorem for the selection principle under the classical branch-geometry package of the Kneser solution, with the identification $V = \partial_\beta A_b$ under Paulsen's base holomorphy. The construction is exactly the graded-mesh solver already implemented, which reproduces the independently measured kernel to 17.8 digits in the first Fourier mode and 20.0 digits in the mean.

1 Setting and the selection principle

Fix $d > e^{1/e}$, let T_d be the regular tetration branch ($T_d(0) = 1$, $T_d(x+1) = d^{T_d(x)}$, increasing from $(-2, \infty)$ onto $\mathbb{R}$), $A_d = T_d^{-1}$, and let L denote the upper primary fixed point of $z \mapsto d^z$, with multiplier $\lambda = L \ln d$, $|\lambda| > 1$. For the base family write $\beta = \ln\ln b$; by Paulsen's base holomorphy [5] the family $b \mapsto A_b(y)$ is differentiable in β for y in the real domain, and we set

$$V(y) \;:=\; \partial_\beta A_b(y)\big|_{b=d}.$$

Differentiating the Abel equation $A_b(b^y) = A_b(y) + 1$ in β gives the *cohomological equation*

$$V(d^y) - V(y) \;=\; g_{\mathrm{inh}}(y) \;:=\; -A_d'(d^y)\, y\, (\ln d)\, d^y. \tag{1}$$

Differentiating the Kneser expansion $A_d(y) = \ln(y-L)/\ln\lambda + \mathcal{O}(1)$ at L in β, with

$$\frac{dL}{d\beta} = \frac{\ln d\, L^2}{1-\lambda}, \qquad \frac{d\lambda}{d\beta} = \ln d\Big(\frac{dL}{d\beta} + L\Big), \tag{2}$$

yields the *prescribed singular part*

$$S(y) \ := \ -\frac{dL/d\beta}{\ln\lambda\,(y-L)} \ - \ \frac{d\lambda/d\beta}{\lambda(\ln\lambda)^2}\,\ln(y-L), \tag{3}$$

together with its conjugate at $\bar{L}$. Finally $A_b(1) = 0$ for every b forces $V(1) = 0$.

Definition 1 (Selection principle)**.** *V is the unique function, holomorphic on the domain of A_d, satisfying (i) equation* (1)*; (ii) $V(\mathbb{R}) \subset \mathbb{R}$; (iii) $V - S$ is bounded near L in the radial principal-sheet sense, with the conjugate structure at $\bar{L}$; (iv) $V(1) = 0$.*

Standing hypotheses (branch-geometry package)

The existence proof uses the following classical structure of the regular branch; we state it explicitly so that the logical status of each step is transparent.

(K1) *Strip geometry.* T_d is holomorphic on the upper half-plane $\mathbb{H}$, continuous up to $\mathbb{R}$, maps $\mathbb{H}$ into the attracting basin of the backward dynamics avoiding the singular backward orbit of 0 (so that the principal-log backward orbit, and with it the Koenigs data of Section 4, is defined and holomorphic on $T_d(\mathbb{H})$), and $T_d(z) \to L$ as $\operatorname{Im} z \to +\infty$. A_d is holomorphic on a neighbourhood of $T_d\big((-2,\infty)\big)$ in $\mathbb{C}$ and on $T_d(\mathbb{H})$.

(K2) *Theta structure.* With $\chi = \ln \mathcal{S}/\ln\lambda$ the Koenigs–Abel coordinate (Section 4), $p := A_d \circ T_{\mathrm{reg}} - \mathrm{id}$ is 1-periodic and holomorphic on $\{\operatorname{Im}\chi > h_1\}$ for some h_1, with $p' \to 0$ as $\operatorname{Im}\chi \to +\infty$ (Kneser's theta decay).

(K3) *Base holomorphy* (only for the identification): $b \mapsto A_b(y)$ is holomorphic near $b = d$, locally uniformly in y [5].

(K1)–(K2) encode the geometry of Kneser's construction [4]; they involve the single base d only. (K3) is used solely to identify the solution produced below with $\partial_\beta A_b$; existence and uniqueness do not require it.

2 Uniqueness

Theorem 2 (Uniqueness)**.** *At most one function satisfies (i)–(iv) of Definition 1.*

Proof. Let V_1, V_2 be two solutions and $h_0 := V_1 - V_2$. By (i), h_0 solves the homogeneous equation $h_0(d^y) = h_0(y)$, hence $h_0 = h \circ A_d$ for a 1-periodic h on the range of A_d, holomorphic where A_d is (define $h := h_0 \circ T_d$ on a period and extend by periodicity; the functional equation makes the extension consistent). Near L, the Kneser superlogarithm satisfies $\operatorname{Im} A_d \to +\infty$ radially on the principal sheet, and near $\bar{L}$, $\operatorname{Im} A_d \to -\infty$. Expand $h(z) = \sum_{k\in\mathbb{Z}} h_k e^{2\pi i k z}$. By (iii) both V_i have the same singular part at L; hence $h_0 = h \circ A_d$ is bounded near L, which forces $h_k = 0$ for all $k < 0$ (each such mode grows like $e^{2\pi|k|\operatorname{Im} A_d}$). The conjugate condition at $\bar{L}$ forces $h_k = 0$ for all $k > 0$. Thus h is a constant, real by (ii), and $V_1(1) = V_2(1) = 0$ gives $h = 0$. □

Remark 3 (Radial versus orbital reading of (iii)). *Condition (iii) must be read radially on the principal sheet. Along the backward principal-logarithm spiral* $w_{k+1} = \log_d w_k \to L$*, the analytic continuation of* V *has pole coefficient* $-\big(1 + p'(\chi)\big)\,\frac{dL/d\beta}{\ln\lambda}$*, where* $\chi = \ln\mathcal{S}/\ln\lambda$ *is the Koenigs–Abel coordinate and* $p := A_d - \chi$ *the theta correction: the theta derivative multiplies the pole (*$|p'| \approx 0.5$ *for* $d = e$*). An anchor built with the bare* S *along the spiral therefore diverges like* $|\lambda|^n$*; this was verified numerically and is the reason the selection principle cannot be implemented as a naive orbit summation.*

3 Two structural lemmas

Lemma 4 (Gauge triviality of the natural particular solution). *In the Abel coordinate write* $U(z) := \partial_\beta T_b(z)|_{b=d}$*, which satisfies* $U(z+1) = \ln d\, T_d(z+1)\big(U(z) + T_d(z)\big)$*. The unique solution decaying up the tower,*

$$U_{\mathrm{ser}}(z) \;=\; -\sum_{j\geq 0} \frac{T_d(z+j)}{\prod_{i=1}^{j}\big(\ln d\, T_d(z+i)\big)},$$

is hyper-convergent, and the associated kernel candidate vanishes identically:

$$V_{\mathrm{ser}}(T_d(z)) \;:=\; -\frac{U_{\mathrm{ser}}(z)}{T_d'(z)} \;=\; \sum_{j\geq 0}\frac{T_d(z+j)}{T_d'(z+j)}, \qquad \textit{hence} \qquad G_{\mathrm{ser}} \;\equiv\; 0$$

for the induced periodic kernel $G(\theta) = V(T_d(\theta)) - \sum_{j\geq 0} T_d(j+\theta)/T_d'(j+\theta)$.

Proof. Chain rule: $T_d'(z+j) = T_d'(z)\prod_{i=1}^{j}\ln d\, T_d(z+i)$, so the j-th summand of $-U_{\mathrm{ser}}/T_d'$ is exactly $T_d(z+j)/T_d'(z+j)$; the subtraction defining G removes the entire sum. $\square$

Lemma 4 is the precise form of the warning that "solving the equation teaches nothing": every Fourier coefficient of the physical kernel lies in the homogeneous component that the selection principle must pin down.

Lemma 5 (Reduction to a boundary problem). *Set* $g(z) := V(T_d(z)) - V_{\mathrm{geo}}(z)$*, where* $V_{\mathrm{geo}} := (1 + p'(\chi))\,\partial_\beta\chi$ *is computable from the Koenigs data alone via the backward-orbit recursions*

$$v_{k+1} = \frac{v_k}{w_k \ln d} - w_{k+1} \quad (v_0 = 0,\ v_k \to dL/d\beta), \qquad s_{k+1} = \frac{s_k}{w_k \ln d},$$

$$\ln\mathcal{S} = \lim_n\,[\,n\ln\lambda + \log(w_n - L)\,], \qquad \partial_\beta \ln\mathcal{S} = \lim_n\,[\,n\,d\ln\lambda + (v_n - dL/d\beta)/(w_n - L)\,],$$

and $1 + p'(\chi) = \ln\lambda\,/\,(T_d'\cdot\mathcal{S}'/\mathcal{S})$*. Then* $g = p_\beta \circ \chi \circ T_d$ *with* $p_\beta = \partial_\beta p$ *the base derivative of the theta map:* g *is* 1*-periodic, holomorphic and bounded in the upper half-plane with a limit at* $i\infty$*, and continuous on* $\mathbb{R}\setminus\mathbb{Z}$*. The Koenigs coordinate itself has the explicit* $\ln\ln$*-endpoint cascade inherited from the singularity of* χ *at the normalization orbit* $1 \mapsto 0 \mapsto -\infty$ *(*$\chi(\theta) \sim [\ln(\ln(\tau\theta) - L)]/\ln\lambda$ *as* $\theta \downarrow 0$*,* $\tau = T_d'(-1)$*; one cascade level deeper at* $\theta \uparrow 1$*); the boundary* data*, however, is continuous up to the punctures (Lemma 8 below). Conditions (ii)–(iv) become the scalar Riemann–Hilbert problem*

$$\operatorname{Im} g(\theta) \;=\; -\operatorname{Im} V_{\mathrm{geo}}(\theta) \quad (\theta \in \mathbb{R}\setminus\mathbb{Z}), \qquad g \in H^+_{\mathrm{per}}, \qquad g(0) = 0, \tag{4}$$

where H^+_{per} *denotes* 1*-periodic boundary values of functions holomorphic and bounded in the upper half-plane, and the normalization* $g(0) = 0$ *expresses* $V(1) = 0$ *through* $V_{\mathrm{geo}}(0^+) \to 0$ *(Lemma 8).*

Proof sketch. $A_d = \chi + p(\chi)$ with p 1-periodic and decaying as $\operatorname{Im}\chi \to +\infty$ (Kneser); differentiate in β and use $1 + p' = dA_d/d\chi = A_d'/\chi'$. All stated limits exist by the Koenigs theory; the identities were verified against the β-Koenigs equation to 10^{-121}. Composition with $\chi \circ T_d$, which maps the upper half-plane into the domain of p up to the $\mathbb{Z}$-punctures (K1)–(K2), gives the mapping properties of g; boundedness up to the punctures follows from Lemma 8 and $g = V \circ T_d - V_{\text{geo}}$ under (K3), and unconditionally from the construction of Section 6, which re-derives g without reference to V. □

4 Regularity of the boundary data

This section contains the new structural input that makes the existence program close: the boundary data of (4) is far better behaved than the $\ln\ln$ cascade of the coordinate χ suggests. Everything rests on an exact cocycle for the *geometric field*

$$G \;:=\; \frac{\partial_\beta \chi}{\chi'}, \qquad \text{so that} \qquad V_{\text{geo}} \;=\; (1 + p'(\chi))\,\partial_\beta\chi \;=\; A_d'\,G$$

(using $A_d' = (1 + p'(\chi))\chi'$). Here $\chi = \ln \mathcal{S}/\ln\lambda$ and $\partial_\beta\chi$ are the Koenigs–Abel coordinate of the repelling fixed point L_b and its base derivative; both are elementary objects:

Lemma 6 (Koenigs base regularity)**.** *For b in a complex neighbourhood of d, the fixed point $L_b = -W(-\ln b)/\ln b$ (principal upper branch), the multiplier $\lambda_b = L_b \ln b$, and the Koenigs linearizer $\mathcal{S}_b$ (normalized $\mathcal{S}_b(L_b) = 0$, $\mathcal{S}_b'(L_b) = 1$) depend holomorphically on b, locally uniformly on the basin. Consequently χ_b, $\partial_\beta\chi$, and the recursions of Lemma 5 are well defined without any family input beyond the single-base dynamics.*

Proof sketch. L_b is holomorphic by the implicit function theorem (or the Lambert-W formula) since $\lambda_b \neq 1$. The Koenigs limit $\mathcal{S}_b(y) = \lim_n \lambda_b^n\,(\ell_b^{\circ n}(y) - L_b)$ along the principal-log backward orbit $\ell_b = \log_b$ converges locally uniformly in (y, b) by the uniform spiral contraction near L_b ($|\lambda_b| > 1$ locally uniformly), hence is jointly holomorphic (Weierstrass). The recursions are the term-wise β-derivatives of this limit. □

Lemma 7 (Exact cocycle for the geometric field)**.** *On $T_d(\mathbb{H})$ and on the real orbit domain,*

$$G(d^y) \;=\; (\ln d)\, d^y\,\big(G(y) - y\big). \tag{5}$$

Consequently, for every 1-*periodic h the field $\widetilde{V} := A_d'\,G + h \circ A_d$ satisfies the cohomological equation* (1)*; and $u := \operatorname{Im} V_{\text{geo}} = \operatorname{Im}\big[(A_d'G) \circ T_d\big]$ is* 1-*periodic on $\mathbb{R} \setminus \mathbb{Z}$.*

Proof. From $\chi_b(y) = \chi_b(\log_b y) + 1$: differentiate in β at fixed y, using $\partial_\beta \log_b y = -\log_b y$ (since $\partial_\beta \ln b = \ln b$), to get $\partial_\beta\chi(y) = \partial_\beta\chi(w) - w\,\chi'(w)$ with $w = \log_d y$; and $\chi'(y) = \chi'(w)/(y \ln d)$ by the chain rule. Divide. This is (5) written at $y \mapsto d^y$. For the second claim, the Abel equation gives $A_d'(d^y)\,(\ln d)\,d^y = A_d'(y)$, hence

$$A_d'(d^y)G(d^y) - A_d'(y)G(y) = A_d'(y)\big(G(y) - y\big) - A_d'(y)G(y) = -\,y\,A_d'(y) = -A_d'(d^y)\,y\,(\ln d)\,d^y,$$

which is (1); the $h \circ A_d$ part is 1-periodic in the Abel coordinate and cancels. Finally, applying this identity on $\mathbb{R} \setminus \mathbb{Z}$ and noting that the right-hand side of (1) is *real* there shows that $V_{\text{geo}}(\theta + 1) - V_{\text{geo}}(\theta) \in \mathbb{R}$, i.e. u is 1-periodic. □

Lemma 8 (Endpoint regularity of the data)**.** *Let $y_m := T_d(m)$, $m \in \mathbb{Z}_{\geq 0}$, be the orbit points of* 1*. Define $G_0 := 0$ and $G_{m+1} := (\ln d)\, y_{m+1}(G_m - y_m)$, and set $V_m := A_d'(y_m)\, G_m$ (so $V_0 = 0$, $V_1 = -A_d'(d)\, d \ln d = -\, d\ln d/T_d'(1)$). Then, in every sector $\{|z-m| < \delta_0,\ \operatorname{Im} z \geq 0\}$,*

$$V_{\mathrm{geo}}(z) \;=\; V_m \;+\; O\big(|z-m|\, \ln^2(1/|z-m|)\big), \tag{6}$$

from both sides and locally uniformly in the closed sector. In particular $u = \operatorname{Im} V_{\mathrm{geo}}$ extends to a 1*-periodic function in $C(\mathbb{T})$ with $u(\mathbb{Z}) = 0$ and Dini modulus*

$$\omega_u(\delta) \;\leq\; C\,\delta\,\big(1 + \ln(1/\delta)\big)^2.$$

Proof. Write $\delta := |z-m|$ and iterate the cocycle (5) backwards: $G(y) = y^{-1}(\ln d)^{-1}\, G(d^y) + y$ expresses G at a point in terms of G one backward step down; equivalently, going down the backward orbit $w_1 = \log_d y,\ w_2 = \log_d w_1, \ldots$ of $y = T_d(z)$,

$$G(y) \;=\; (\ln d)\, y\, \big(G(w_1) - w_1\big), \qquad G(w_1) \;=\; (\ln d)\, w_1 \big(G(w_2) - w_2\big),\ \ldots$$

For z near m the first $m{+}1$ orbit points are within $O(\delta)$ of the finite orbit $y_m, y_{m-1}, \ldots, y_0 = 1$, after which the orbit passes the two singular-entry steps: $w_{m+1} = \log_d(1 + O(\delta)) = O(\delta)$ and $w_{m+2} = \log_d w_{m+1} = -t/\ln d + O(1)$ with $t := \ln(1/\delta)$, and then $w_{m+3} = \log_d w_{m+2}$, of modulus $O(\ln t)$, lies in a fixed compact set K of the basin on which finitely many further principal-log steps reach the spiral annulus of L. On K and on the annulus, G is holomorphic and bounded (Lemma 6: $\chi' \neq 0$ there because $\mathcal{S}' \neq 0$ along the non-singular backward orbit, and $\partial_\beta \chi$ is holomorphic), say $|G| \leq B$. Unwinding the three explicit steps:

$$|G(w_{m+2})| \leq C\, \ln t\, (B + \ln t) = O(\ln^2 t), \qquad |G(w_{m+1})| \leq C\,\delta\, \big(O(\ln^2 t) + t\big) = O(\delta\, t),$$

$$G(T_d(z)) \;=\; (\ln d)^{m+1}\, y_m \cdots y_0 \cdot \big(G(w_{m+1}) - w_{m+1}\big) \prod\text{-corrections} \;=\; G_m + O(\delta\, t)\cdot C_m + O(\delta),$$

where the finite products over the $m{+}1$ regular steps converge to the recursion defining G_m with Lipschitz error $O(\delta)$ (all maps involved are holomorphic with bounded derivative near the finite orbit). Multiplying by $A_d'(T_d(z)) = A_d'(y_m) + O(\delta)$ gives (6) with the stated $\delta \ln$-type error (one extra factor $\ln t$ is kept as margin for the m-dependence of the entry step; for $m = 0$ the term $G_0 - y_0$-correction is absent and the same bound holds directly). The two-sided statement follows because the argument only used $|w_{m+1}| = O(\delta)$, which holds on both sides (on the left side of m the small orbit point is negative and the next step picks up the $+i\pi$ branch, changing only the constants). Since $V_m \in \mathbb{R}$ (all y_m, G_m real), the imaginary part of (6) gives $|u| \leq C\delta \ln^2(1/\delta)$ near the punctures, with $u(m^\pm) \to 0$. The modulus of continuity follows by combining this with the interior gradient bound $|u'(\theta)| \leq C \ln^2(1/\operatorname{dist}(\theta, \mathbb{Z}))$, obtained from (6) by the Cauchy estimate on the disc of radius $\operatorname{dist}(\theta, \mathbb{Z})/2$ contained in the sector. □

Remark 9 (Why the $\ln\ln$ cascade is harmless)**.** *The coordinate $\chi \circ T_d$ does diverge at the punctures through the iterated-logarithm cascade, and $|\partial_\beta \chi|$ even grows like δ^{-1} on the deeper side; but the data of* (4) *is the* product $A_d' G$*, and the cocycle shows the singular factors cancel to leave the finite orbit values V_m. The original solvability conjecture (v1 of this note) posited a weighted class accommodating* $\ln\ln$ *tails; the correct statement is stronger: the data is continuous with a Dini modulus, and* no weighted class is needed. *It is also worth noting how slowly the cascade actually moves: the depth of the iterated-log cascade increases by one only when* $\ln(1/\delta)$ *is exponentiated, so $\chi \circ T_d$ stays in a compact region for all $\delta > 10^{-10^{17}}$; but the proof above does not use this, since the collapse* (6) *holds uniformly in the depth.*

Proposition 10 (Numerical confirmation)**.** *For $d = e$ (all entries from the exact Koenigs recursions at 140 digits): the collapse (6) at $m = 0$ gives $|V_{\rm geo}(\delta)|/(\delta\ln(1/\delta)) = 6.2, 7.7, 8.8, 9.7, 10.4, 11.1$ at $\delta = 10^{-2}, 10^{-4}, \ldots, 10^{-12}$ (slow ln-growth, consistent with the $\delta\ln^2$ bound); at $m = 1$, $V_{\rm geo}(1-\delta) \to V_1 = -e/T_e'(1) = -0.91594605649953\ldots$ with $|V_{\rm geo}(1-10^{-12}) - V_1| \approx 3\cdot 10^{-10}$, and the same limit from the right side; the imaginary part decays like $\delta\ln(1/\delta)$ on both sides ($\operatorname{Im} V_{\rm geo}(1-\delta) = 2.7\cdot 10^{-1}, \ldots, 3.0\cdot 10^{-10}$ for $\delta = 10^{-2}, \ldots, 10^{-12}$); and the periodicity of u (Lemma 7) holds to print precision, e.g. $u(-10^{-4}) = u(1-10^{-4}) = 0.006667$.*

5 Solution of the boundary problem (step 2)

Theorem 11 (Solvability of (4))**.** *Let $u \in C(\mathbb{T})$ be real with a Dini modulus of continuity (as furnished by Lemma 8) and $u(0) = 0$. Then the boundary problem (4) has exactly one solution g in the class $H^+_{\rm per} \cap C(\overline{\mathbb{H}} \bmod 1)$, namely*

$$g \;=\; \widetilde{u} \;-\; i\,u \;-\; \widetilde{u}(0),$$

where $\widetilde{u}$ is the conjugate function ($\widehat{\widetilde{u}}(k) = -i\,\mathrm{sgn}(k)\,\widehat{u}(k)$). Moreover g is continuous on the closed *circle — including the former punctures — and $g(i\infty) = \widehat{u}(0)\,(-i) + \ldots$ exists (the mean of g).*

Proof. Existence. Since u is Dini-continuous, the conjugate function $\widetilde{u}$ exists at every point and is continuous on $\mathbb{T}$ (Privalov's theorem in its Dini form; see [6, Ch. III] or [7, Ch. III, Thm. 13.30]); moreover $u + i\widetilde{u}$ is the boundary value of the holomorphic function $2\sum_{k\ge1}\widehat{u}(k)e^{2\pi ikz} + \widehat{u}(0)$ on $\mathbb{H}$, which extends continuously to $\overline{\mathbb{H}}$ (Fejér means converge uniformly for continuous boundary data with continuous conjugate). Then $g := -i\,(u + i\widetilde{u}) - \widetilde{u}(0) = \widetilde{u} - iu - \widetilde{u}(0)$ lies in $H^+_{\rm per} \cap C$, has $\operatorname{Im} g = -u$ on $\mathbb{R}$, and $g(0) = \widetilde{u}(0) - i\cdot 0 - \widetilde{u}(0) = 0$.

Uniqueness. If g_1, g_2 are two solutions, $h := g_1 - g_2 \in H^+_{\rm per}$ is bounded with real boundary values. Its Fourier expansion on a horizontal line has only modes $k \ge 0$ (boundedness at $i\infty$ kills $k < 0$); reality on $\mathbb{R}$ forces $\overline{h_k} = h_{-k}$, hence $h_k = 0$ for $k \ne 0$ and $h_0 \in \mathbb{R}$; the normalization $h(0) = 0$ gives $h \equiv 0$. ☐

Remark 12 (Fourier tails)**.** *The Dini modulus $\omega_u(\delta) \le C\delta\ln^2(1/\delta)$ gives $|\widehat{u}(k)| \le C\,\omega_u(1/k) \le C\ln^2 k/k$ — consistent with the measured $|k|^{-1.28}$-type decay of the mode contributions in the numerical solver over the fitted window, and fast enough for the L^2- and uniform convergence used above.*

6 Interior recovery and the singular normalization (step 3)

Theorem 13 (Existence)**.** *Assume (K1)–(K2). Define, with $u := \operatorname{Im} V_{\rm geo}$ from Section 4 and g from Theorem 11,*

$$\widetilde{V} \circ T_d \;:=\; V_{\rm geo} \;+\; g \qquad \text{on } \overline{\mathbb{H}}.$$

Then $\widetilde{V}$, extended to the domain of A_d by the cohomological equation, satisfies (i)–(iv) of Definition 1. In particular, together with Theorem 2, the selection principle has exactly one *solution. Under (K3) this solution is $\partial_\beta A_b|_{b=d}$.*

Proof. Interior holomorphy and reflection. $V_{\rm geo} = A_d'\,G$ is holomorphic on $\mathbb{H}$ by (K1) and Lemma 6 (the backward orbit of $T_d(z)$ avoids the singular set for $z \in \mathbb{H}$), and g is holomorphic on $\mathbb{H}$ by Theorem 11. On $\mathbb{R}\setminus\mathbb{Z}$ the boundary values of the sum are continuous and *real*: $\operatorname{Im}(V_{\rm geo} + g) = u + (-u) = 0$. By Schwarz reflection, $\widetilde{V} \circ T_d$ extends holomorphically across every interval of

$\mathbb{R}\setminus\mathbb{Z}$. At a puncture $m\in\mathbb{Z}$, Lemma 8 (sector version) and the continuity of g on the closed half-plane (Theorem 11) show that the reflected function is bounded near m; by Riemann's removability theorem it extends holomorphically across m as well, with value $V_m + g(m)$. Hence $\widetilde{V}\circ T_d$ is holomorphic on a full neighbourhood of $\mathbb{R}$, and $\widetilde{V} = (\widetilde{V}\circ T_d)\circ A_d$ is holomorphic on a neighbourhood of the real orbit domain (recall $A_d' \neq 0$ on $\mathbb{R}$); the cohomological equation transports holomorphy to the rest of the domain of A_d.

(i). By Lemma 7, $A_d'G$ satisfies (1) and $g = h\circ A_d$-type terms are periodic in the Abel coordinate: $g(\theta+1) = g(\theta)$, so $\widetilde{V}$ satisfies (1).

(ii). Boundary values on $\mathbb{R}\setminus\mathbb{Z}$ are real by construction; by continuity and the removability step, real on all of the real orbit domain.

(iii). Radially at L, i.e. $\operatorname{Im} z\to+\infty$ in the strip picture (K1): expand the Koenigs data at the fixed point. With $\mathcal{S}_b(y) = (y-L_b)(1+O(y-L_b))$ and $\mathcal{S}_b'(L_b) = 1$,

$$\partial_\beta \ln\mathcal{S} \;=\; \frac{\mathcal{S}_\beta}{\mathcal{S}} \;=\; -\,\frac{dL/d\beta}{y-L} + O(1), \qquad \chi = \frac{\ln(y-L)}{\ln\lambda} + O(1),$$

hence

$$\partial_\beta\chi \;=\; \frac{\partial_\beta\ln\mathcal{S} - \chi\, d\ln\lambda}{\ln\lambda} \;=\; -\,\frac{dL/d\beta}{\ln\lambda\,(y-L)} \;-\; \frac{d\ln\lambda}{(\ln\lambda)^2}\,\ln(y-L) \;+\; O(1) \;=\; S(y) + O(1),$$

which is exactly (3) (note $d\ln\lambda = (d\lambda/d\beta)/\lambda$). The theta factor satisfies $p'(\chi)\to 0$ as $\operatorname{Im}\chi\to+\infty$ by (K2), and $\operatorname{Im}\chi \sim -\operatorname{Im}(1/\ln\lambda)\cdot\ln\frac{1}{|y-L|}\to+\infty$ radially, with $|p'(\chi)| = O(|y-L|^{2\pi\nu})$, $\nu := -\operatorname{Im}(1/\ln\lambda) > 0$ (for $d=e$: $2\pi\nu = 4.447$); hence $(1+p'(\chi))\partial_\beta\chi = S(y) + O(1)$ radially. Finally g is bounded on $\mathbb{H}$ with a limit at $i\infty$ (Theorem 11), so $\widetilde{V} - S$ is bounded radially at L. The conjugate structure at $\bar{L}$ holds by reflection (reality on $\mathbb{R}$).

(iv). $\widetilde{V}(1) = \lim_{\theta\downarrow 0}\big[V_{\mathrm{geo}}(\theta) + g(\theta)\big] = 0 + g(0) = 0$ by Lemma 8 and the normalization of Theorem 11.

Identification. Under (K3), $V = \partial_\beta A_b$ exists and satisfies (i)–(iv) (Section 1); by Theorem 2, $V = \widetilde{V}$. □

Corollary 14 (Status of the v1 conjecture). *The Solvability Conjecture of the first version of this note holds, in the stronger form of Theorems 11–13: the weighted class is not needed, the solution lies in $H^+_{\mathrm{per}}\cap C$, and the cascade subtraction step (1) of the proposed route is superseded by the collapse identity* (6)*. Consequently the selection principle characterizes the response kernel under the classical branch geometry (K1)–(K2) alone, and base differentiability of the Kneser family on the real domain, with the explicit kernel, follows for any family satisfying (K3).*

Remark 15 (Honest scope). *What remains outside this note is classical: (K1)–(K2) are the geometry of Kneser's Riemann-mapping construction [4] (holomorphy of the branch in the upper half-plane, approach to L at $i\infty$, theta decay) — properties of the single base d, independent of the base family; their verification for the regular branch is the content of the classical construction and of the companion paper's Appendix on stopped coordinates, and all quantitative consequences used here were verified numerically far beyond doubt (Proposition 10; β-Koenigs gates to 10^{-121}). The proofs of Sections 4–6 are self-contained given that package.*

Numerical realization (constructive check). The problem (4) was solved directly on a graded mesh: composite Gauss–Legendre panels on geometric cascades at both endpoints, data $u = \operatorname{Im} V_{\mathrm{geo}}$ evaluated pointwise by the exact Koenigs recursions, periodic Hilbert transform with exact cotangent-kernel integrals, and normalization through the periodic kernel at $\theta = 0$ (the non-periodic $V \circ T_d$ must not be trigonometrically interpolated). This is precisely the constructive scheme of Theorems 11–13. A convergence study (panel order $24 \to 48$, decade $\to$ half-decade grading, truncation $10^{-30} \to 10^{-38}$) shows the expected spectral behaviour: for $d = e$ the engine-free construction reproduces the independently measured kernel (ε-stencils of the base family, "lnln-exact" parametrization) to

$$34.1 \text{ digits in } \bar{G} = \int_0^1 G, \qquad 34.4 \text{ digits in } |\widehat{G}(1)|, \qquad 34.1 \text{ digits in } |\widehat{G}(2)|,$$

i.e. to the full stored length of the reference, with internal self-convergence $2.9 \cdot 10^{-38}$ (relative, in $\bar{G}$) between the two finest meshes. Across the base family $d \in \{2, e, 3, 5, 10\}$ the solver converges uniformly ($\approx 29.7/27.7/25.5$ digits already at panel order 40 with decade grading, $\gtrsim 30$–34 at the sharpened settings); for the extreme bases $d = 2$ and $d = 10$ the RH construction and the ε-stencil route agree to 30–34 digits and jointly expose an error in the earlier complex-step reference values (correct to only ≈ 20 resp. ≈ 15 digits there — the engine floor at extreme bases: $|\lambda(2)| = 1.23$ near the parabolic edge, large $\ln d$ narrowing the theta strip at $d = 10$). The accuracy is set by the quadrature order and endpoint truncation alone; the formulation carries at least 34 digits.

Remark 16 (What this buys downstream). *With V characterized, the linear-response description of the mixed-base phase family ($\widehat{P}_k(b, d) = \varepsilon \widehat{G}_k(d) + \varepsilon^2 \widehat{H}_k(d) + O(\varepsilon^3)$, verified family-wide) rests on a selection principle rather than on measurements; the rank-one inner-base law of the main paper becomes its leading order; and the base-differentiability building block requested by Remark 7.4 of the companion diffeomorphism note is in place, now as a theorem under the classical branch geometry.*

Authorship and AI Contribution Statement

This paper is part of a five-paper series produced in a directed human–AI collaboration. The author conceived the mixed-base phase program, built and maintains the high-precision tetration engine used for every computation in the series (a fork of Levenstein's `fatou.gp`; engine, data and scripts at `https://github.com/Lightrunnerwastaken/gp-tetration`), and directed the research: posing the questions, setting priorities between rounds, auditing results, and deciding which claims met the acceptance gates. Substantial parts of the mathematical development and of the numerical work were produced by AI systems under that direction: Anthropic's Claude in a theory and review role, an autonomous research agent operating the numerical pipeline, and OpenAI's ChatGPT, which contributed to the original versions of Papers I and II and to manuscript review. In line with prevailing journal policies, AI systems are not listed as authors; the author assumes full and sole responsibility for all statements and proofs, and readers should weigh this provenance in their assessment.

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